Data Science & Artificial Intelligence

Machine Learning

KNN

DPP-01

Discussion Notes

By - Siddharth Sabharwal Sir

[NAT]

#Q. Consider a set of five training examples given as $((x_i, y_i), c_i)$ values, where x_i, and y_i are the two attribute values (positive integers) and c_i is the binary class label:

1. $((1, 1), -1) \rightarrow 2+5=7$
2. $((1, 7), +1) \rightarrow 2+1=3$
3. $((3, 3), +1) \rightarrow 0+3=3$
4. $((5, 4), -1) \rightarrow 2+2=4$
5. $((2, 5), -1) \rightarrow 1+1=2$

Classify a test example at coordinates (3, 6) using a k-NN classifier with k = 3 and Manhattan distance.

Sol Test point (3,6), Now we find manhattan distance from each training point

So min distance is for 2, 3, 5 $\rightarrow$ +1, +1, -1 $\Rightarrow$ +1

[NAT]

#Q. Predicting Movie Genre 1NN

	IMDB Rating	Duration	Genre
1	8.0 (Mission Impossible)	160	Action
2	6.2(Gadar 2)	170	Action
3	7.2 (Rocky & Rani)	168	Comedy
4	8.2 (OMG 2)	155	Comedy

Test point 7.4, 114

$\rightarrow \sqrt{.6^2 + 46^2}$

$\rightarrow \sqrt{1.2^2 + 56^2}$

$\rightarrow \sqrt{.2^2 + 54^2}$

$\rightarrow \sqrt{.8^2 + 41^2}$

The genre of "Barbie" movie with IMDb rating 7.4 and duration 114.

Now predict the genre of.... Barbie

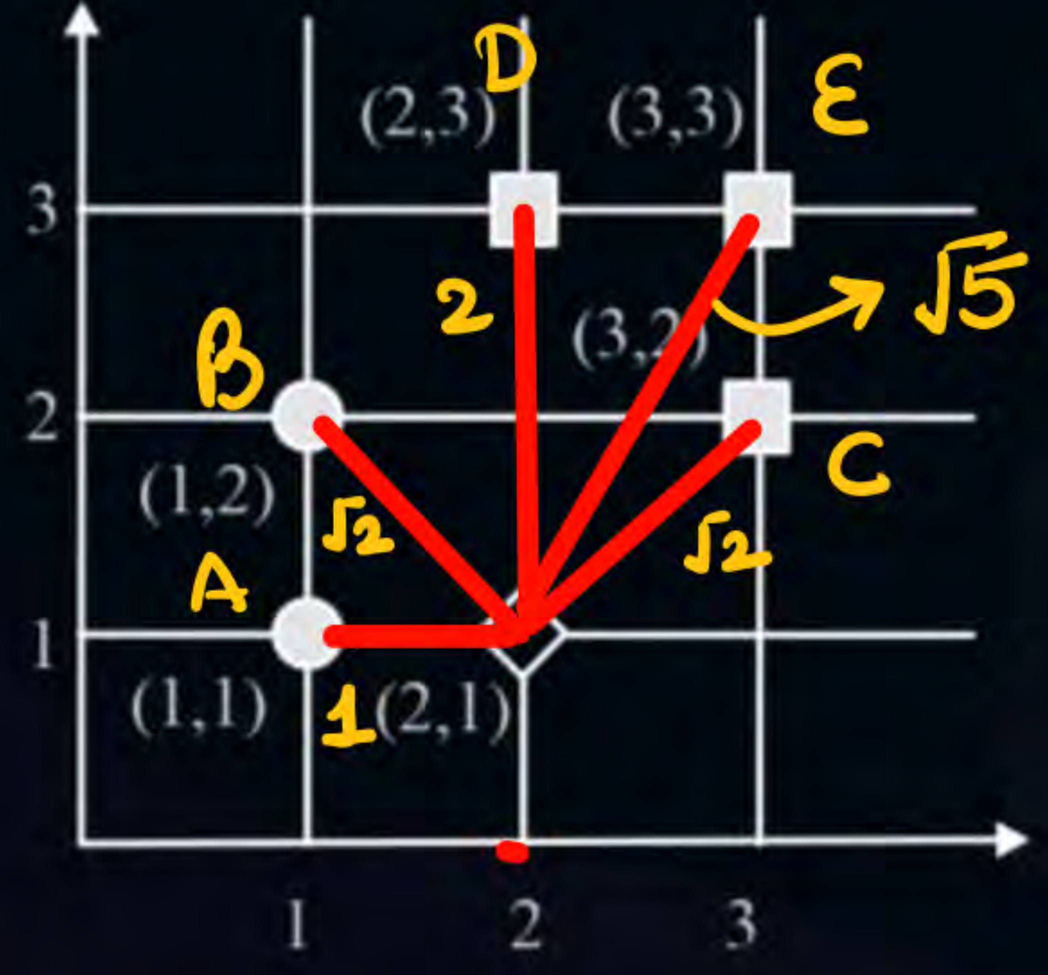
Sol Test point 7.4, 114

So 4th point is closest $\rightarrow$ 1NN $y_{Barbie} = y_{omg2}$

$= \text{Comedy}.$

[NAT]

#Q. Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in nearest neighbor algorithm for which the diamond(◆) shaped data point is assigned the label square is



Sol.

- 1NN → A → ●
- 3NN → A, B, C → ●
  ● ● ■
- 5NN → ABCDE
  ● ● ■ ■ ■ → ■

K=5 → ■

[MCQ]

#Q. Which of the following are true for the k-nearest neighbour (k-NN) algorithm?

• Sol → • as K inc overfit dec → bias Training error inc

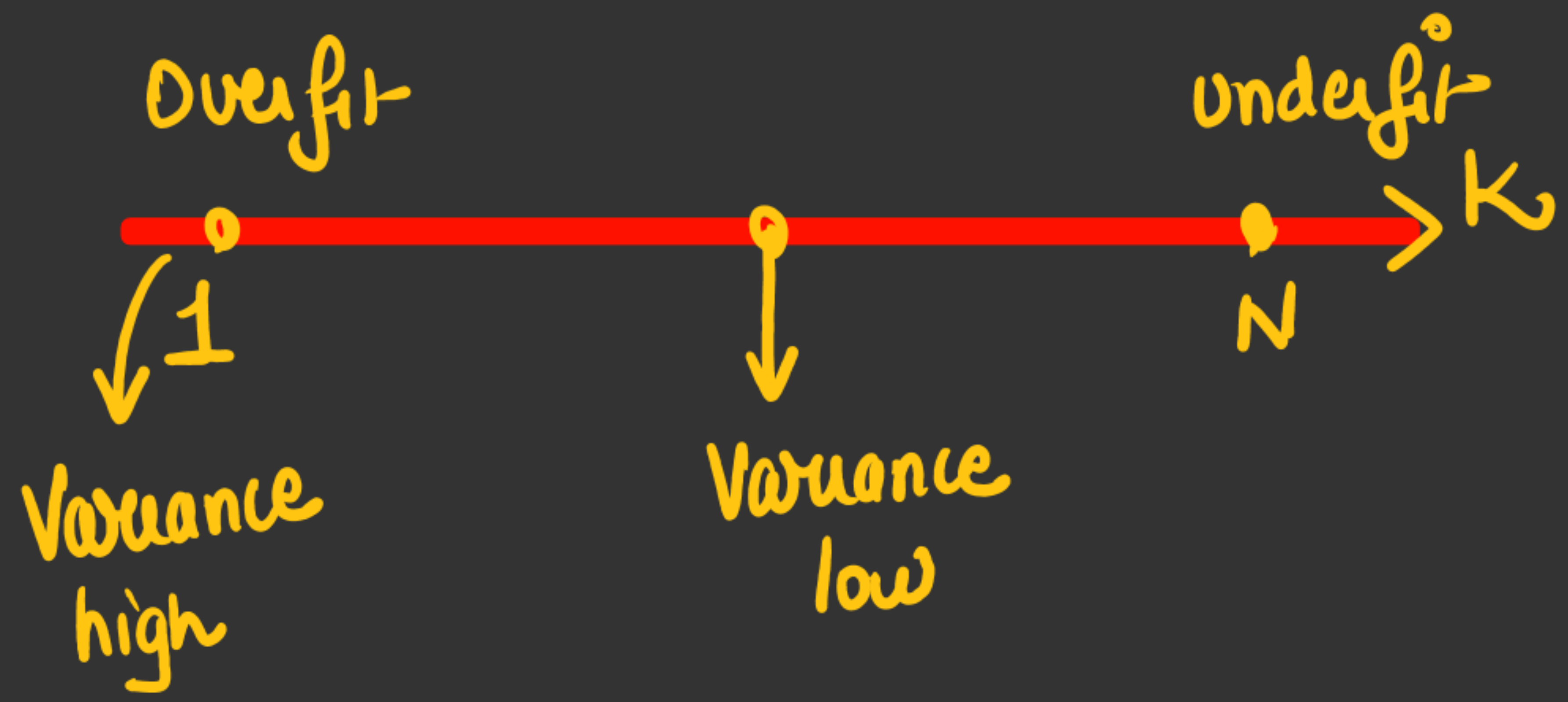
• when K is small : overfit → complex boundary

A ✓ K-NN can be used for both classification and regression.

B ✓ A's k increases, the bias usually increases.

C ✗ The decision boundary looks smoother with smaller values of k.

D ✗ As k increases, the variance usually increases.


Overfit
underfit
K
1
N
Variance
high
Variance
low

[NAT]

#Q.

Sl. No.	Swim	Fly	Crawl	Class
1	Fast	No	No	Fish
2	Fast	No	Yes	Animal
3	Slow	No	No	Animal
4	No	No	No	Animal
5	No	Short	No	Bird
6	No	Short	No	Bird
7	No	Rarely	No	Animal
8	Slow	No	Yes	Animal
9	Slow	No	No	Fish
10	Slow	No	Yes	Fish
11	No	Long	No	Bird
12	Fast	No	No	Bird

Slow	Rarely	No
1	1	0 → 2
1	1	1 → 3
0	1	0 → 1
1	1	0 → 2
1	1	0 → 2
1	1	0 → 2
1	0	0 → 1
0	1	1 → 2
0	1	0 → 1
0	1	1 → 2
1	1	0 → 2
1	1	0 → 2

The test instance (Slow, Rarely, No) 3NN → find class of Test point

Sol dimension are Categorical in Nature → Hamming distance

Hamming distance → dimension equal distance = 0

" not " distance = 1

point 2, 7, 9 are NN → Animal

Animal

Fish

So Animal will be assigned.

[MCQ]

#Q. In k-NN, what is the effect of using weighted voting for classification?

- So weighted voting → • Close neighbour are given more Importance
  • Outlier effect reduced, overfit reduce ⇓ Bias inc

A. It gives equal importance to all neighbors

B. ✓ It assigns more importance to closer neighbors

C. ✓ It reduces the effect of outliers

D. ✓ It increases the model's bias

[MCQ]

#Q. What is the training error (fraction labeled wrong) in the above picture using 1-nearest neighbor and the L_1 distance norm be-tween points? If there is more than one equally close nearest neighbor, use the majority label of all the equally close nearest neighbors. Please make sure you really are computing training error, not testing error.

- L_1 distance : Manhattan distance
- L_2 " : Eucledian dist

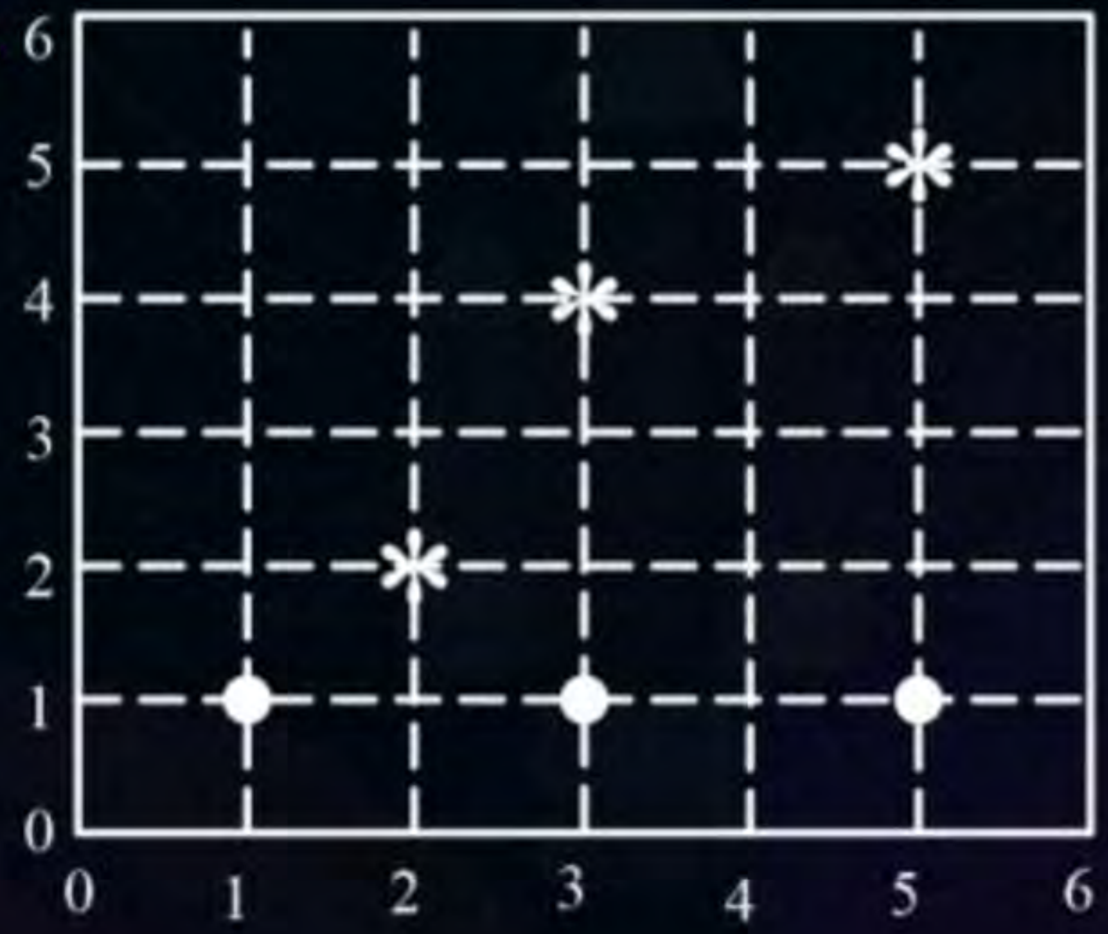
A. 0

B. 1/6

C. 2/6

D. 4/6

- Training error @ 1NN → 0



[NAT]

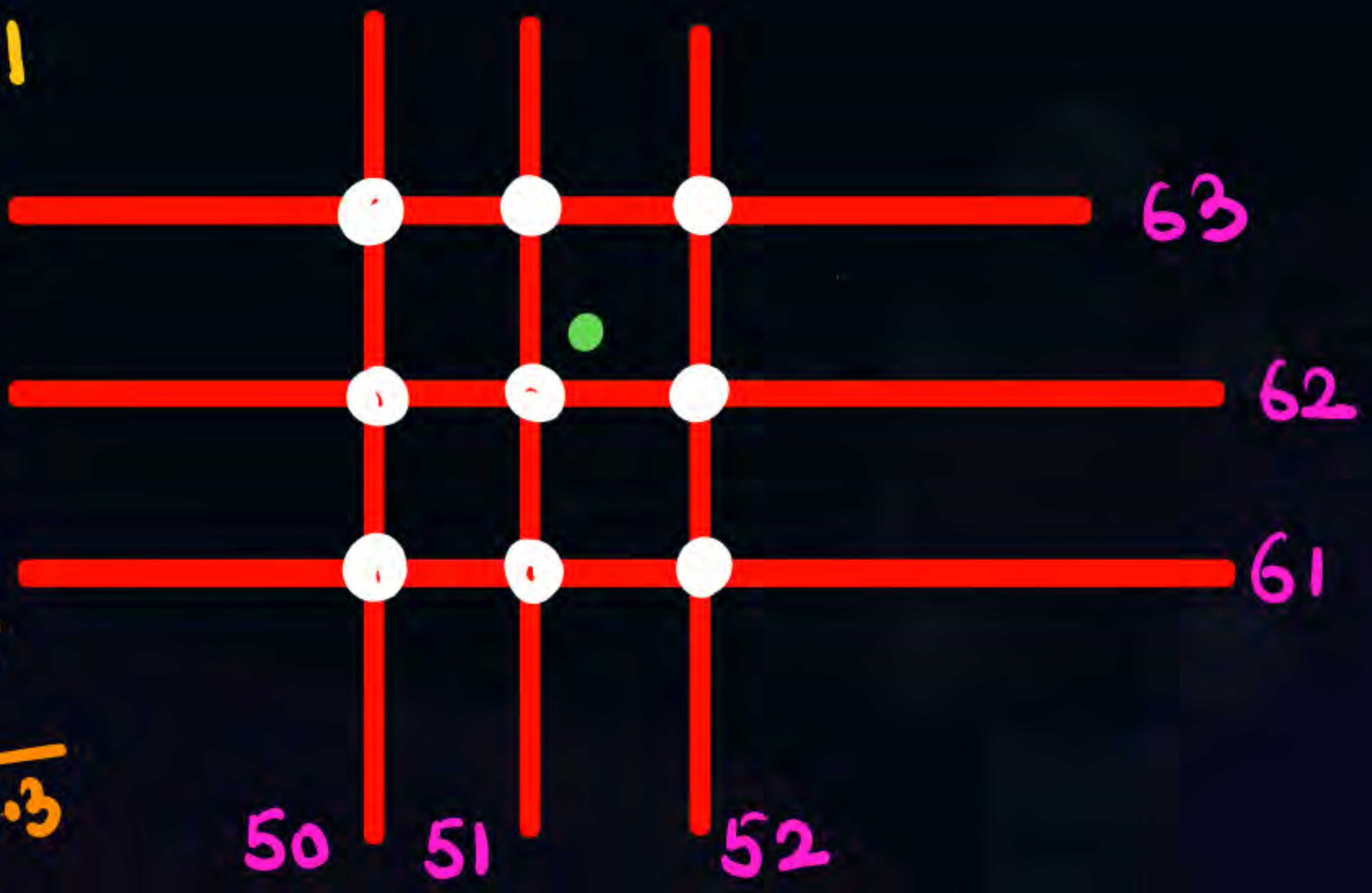
#Q. Consider a 2D dataset where training points are on the integer grid (x_1, x_2), and both dimensions x_1, and x_2 range from 1 to 100 (inclusive). The binary label for point (x_1, x_2) is $(-1)^{(x1+x2)}$ What is the label for test point (51.3, 62.1) with a 3-NN classifier?

Sol 2D data 51.3, 62.1

Closest NN can be

- 51, 62 → $\sqrt{.3^2 + .1^2} = \sqrt{.1}$
- 51, 63 → $\sqrt{.3^2 + .9^2} = \sqrt{.9}$
- 52, 62 → $\sqrt{.7^2 + .1^2} = \sqrt{.5}$

52, 63 → $\sqrt{.7^2 + .9^2} = \sqrt{1.3}$



So $(51,62)$, $(51,63)$, $(52,62)$ are NN

$$\begin{array}{ccc} (51,62) & (51,63) & (52,62) \\ \downarrow & \downarrow & \downarrow \\ (-1)^{odd} & (-1)^{even} & (-1)^{even} \\ (-1) & 1 & 1 \end{array}$$

$1 \longrightarrow$ Class 1.

[MCQ]

#Q. Which of the following statements is/are true in case of k-NN?

⇒ (all are true)

A. In case of very large value of k, we may include points from other classes into the neighborhood.

B. In case of too small value of k, the algorithm is very sensitive to noise.

C. K-NN is non-parametric algorithm

D. k-NN algorithm does more computation on test time rather than train time.

[MCQ]

#Q. Consider the following dataset for a 3-nearest neighbor (3NN) classification problem:

Point	Coordinates	Class Label
A1	(2, 10)	C2
A2	(2, 6)	C1
A3	(11, 11)	C3
A4	(6, 9)	C2
A5	(6, 5)	C1
A6	(1, 2)	C1
A7	(5, 10)	C2
A8	(4, 9)	C2

A. C1

B. C2

C. C3

D. Can't Predict

Suppose we have a point P=(5,7) for which we want to determine the class label using the K-nearest neighbor (KNN) algorithm. Which class does the P belongs?

Point	Coordinates	Class Label
A1	(2, 10)	C2
A2	(2, 6)	C1
A3	(11, 11)	C3
A4	(6, 9)	C2
A5	(6, 5)	C1
A6	(1, 2)	C1
A7	(5, 10)	C2
A8	(4, 9)	C2

Test Point 5, 7

$\rightarrow \sqrt{3^2+3^2}$

$\rightarrow \sqrt{3^2+1^2}$

$\rightarrow \sqrt{6^2+4^2}$

$\rightarrow \sqrt{1^2+2^2}$ ✓

$\rightarrow \sqrt{1^2+2^2}$ ✓

$\rightarrow \sqrt{4^2+5^2}$

$\rightarrow \sqrt{3^2}$

$\rightarrow \sqrt{1^2+2^2}$ ✓

So 3NN $\rightarrow$ $\underset{C_2}{A4}$, $\underset{C_1}{A5}$, $\underset{C_2}{A8}$ $\Rightarrow$ C_2 ✓

[MCQ]

#Q. Suppose you have a dataset with the following points in a two-dimensional space:

{(2,3), (3,5), (5,2), (8,8), (10,7), (12,10)}

Each point belongs to either Class A or Class B. The classes are labeled as follows:

- Class A: (2,3), (3,5), (5,2)
- Class B: (8,8), (10, 7), (12,10)

Now, a new point (7, 6) is given. Apply the k-Nearest Neighbors algorithm with k = 3 to predict the class of this new point.

A. Class A

B. Class B

C. Can't Decide

D. Both Class A and B

Sol

- Class A: (2,3), (3,5), (5,2)
- Class B: (8,8), (10, 7), (12,10)

Distance from (7,6)

$\Downarrow$

3NN $\rightarrow$ A, B, B

$\Downarrow$

B

$2,3 \longrightarrow \sqrt{5^2+3^2} = \sqrt{34}$

A $\cdot$ $3,5 \longrightarrow \sqrt{4^2+1^2} = \sqrt{17}$ ✓

$5,2 \longrightarrow \sqrt{2^2+4^2} = \sqrt{20}$

B $\cdot$ $8,8 \longrightarrow \sqrt{1^2+2^2} = \sqrt{5}$ ✓

B $\cdot$ $10,7 \longrightarrow \sqrt{3^2+1^2} = \sqrt{10}$ ✓

$12,10 \longrightarrow \sqrt{5^2+4^2} = \sqrt{41}$

[MCQ]

#Q. In a KNN classifier with K = 5, if the majority class among the 5 nearest neighbors is class A, what is the predicted class for a new data point?

A. Class A

B. Class B

C. Cannot be determined

D. Depends on the distance metric

[MCQ]

#Q. Match List-I with List-Il and select the correct answer using the codes given below the lists

List-I	List-II
A. K-Nearest Neighbors (KNN)	1. Discriminative Model
B. Forward selection	2. Dimensionality Reduction
C. Naive Bayes classifier	3. Generative Model

Code

	A	B	C
A	1	2	3
B	1	3	2
C	2	3	1
D	2	1	3

[MCQ]

#Q. Which of the following statements is/are true?

A. In k-Nearest Neighbors (k-NN), the classification is based on the majority vote among the k nearest neighbors.

B. Lasso Regression includes all features in the final model, regardless of their relevance.

C. Principal Component Analysis (PCA) uses Euclidean distance to perform dimensionality reduction.

D. Logistic Regression estimates the probability of a binary outcome using a linear function.

[MCQ]

#Q. Which of the following statements are true?

A ✗ Training a k-nearest-neighbors classifier takes more computational time than ap-plying it.

B ✓ The more training examples, the more accurate the prediction of a k-nearest-neighbors.

C ✗ k-nearest-neighbors cannot be used for regression.

D ✓ A k-nearest-neighbors is sensitive to outliers.

[NAT]

#Q. Consider these training examples:

x	y
(0,0)	1
(1,2)	1
(2,3)	0
(5,3)	0

2,1

→ 2+1=3 ✓

→ 1+1=2 ✓

→ 0+2=2 ✓

→ 3+2=5

How does 3-NN (three nearest neighbors) classify (2,1) using LI distance?

$\Downarrow$

1,1,0 → 1 ✓

[MCQ]

#Q. Imagine you are using a k-Nearest Neighbor classifier on a dataset with lots of noise. You want your classifier to be less sensitive to the noise. Which of the following is likely to help and with what side effect?

less overfit

A. Increase the value of k→ Increase in prediction time

B. Decrease the value of k→ Increase in prediction time

C. Increase the value of k → Decrease in prediction time

D. Decrease the value of k → Decrease in prediction time

[MCQ]

#Q. A classifier maps a feature vector x in a vector space X to a discrete label y. Two classifiers C_1 and C_2 are considered equal if, for all $x \in X$, $C_1(x) = C_2(x)$, i.e., the output labels produced by both classifiers are identical for every possible input in the feature space.

Suppose we have a training dataset S and two 1-Nearest Neighbor (1-NN) classifiers, C_1 and C_2, trained on S.

The equality of the classifiers is analyzed in the following cases: owing cases:

- Case (a): C_1 uses the L1-distance, and C_2 uses the L2-distance.
- Case (b) C_1 uses the L2-distance, and C_2 uses the square of the L2-distance.

Are C_1 and C_2 equal in the two cases described above?

Sol Case b we know that if $\sqrt{a}$ is min then a is also min

A. C_1 is equal to C_2 in both case (a) and case (b).

B. C_1 is not equal to C_2 in case (a), but C_1 is equal to C_2 in case (b).

C. C_1 is equal to C_2 in case (a), but C_1 is not equal to C_2 in case (b).

D. C_1 is not equal to C_2 in either case (a) or case (b).

Ex

A(0,0)

B(1,1)

C (1.5, .5)

Test point (1,0)

$$\left.\begin{array}{l} \ell 1 \text{ distance} \quad A \to 1 \\ \qquad\qquad\quad B \to 1 \\ \qquad\qquad\quad C \to .5 + .5 = 1 \end{array}\right\} \text{all 3 equal}$$

$\ell 2$ distance $A = \sqrt{1^2} = 1$

$B = \sqrt{1^2} = 1$

$C = \sqrt{.5^2 + .5^2} = \sqrt{.5}$ ✓

THANK - YOU